MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – TEST (SET - C)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 25**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

MATHS MANTHAN | $(5 \times 1 = 5$ Marks)

Q1. The length of the shadow of a tower on the plane ground is $\sqrt{3}$ times the height of the tower. The angle of elevation of the Sun is: [1]

(A) $30°$ (B) $45°$ (C) $60°$ (D) $90°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $\tan\theta = \dfrac{h}{\sqrt{3}h} = \dfrac{1}{\sqrt{3}} \implies \theta = \boxed{30°}$.

Q2. Raina is 1.5 m tall. At an instant, his shadow is 1.8 m long. At the same time, the shadow of a tree is 9 m long. The height of the tree is: [1]

(A) 3 m (B) 4.5 m (C) 7.5 m (D) 10.8 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: By proportional shadows at the same instant: $\dfrac{H}{9} = \dfrac{1.5}{1.8} = \dfrac{5}{6} \implies H = 9 \times \dfrac{5}{6} = \boxed{7.5 \text{ m}}$.

Q3. A peacock sitting on the top of a tree of height 10 m observes a snake moving on the ground. If the snake is $10\sqrt{3}$ m away from the base of the tree, then the angle of depression of the snake from the eye of the peacock is: [1]

(A) $30°$ (B) $45°$ (C) $60°$ (D) $90°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $\tan\theta = \dfrac{10}{10\sqrt{3}} = \dfrac{1}{\sqrt{3}} = \tan 30° \implies \theta = \boxed{30°}$.

Q4. From a point on the ground, which is 60 m away from the foot of a vertical tower, the angle of elevation of the top of the tower is found to be 30°. The height of the tower is: [1]

(A) $20\sqrt{3}$ m (B) $30\sqrt{3}$ m (C) $60\sqrt{3}$ m (D) 20 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $h = 60\tan 30° = \dfrac{60}{\sqrt{3}} = \boxed{20\sqrt{3}\text{ m}}$.

Q5. **Assertion (A):** A ladder leaning against a wall stands at a horizontal distance of 6 m from the wall. If the height of the wall up to which the ladder reaches is 8 m, then the length of the ladder is 10 m.
Reason (R): The ladder makes an angle 60° with the ground. [1]

(A) Both (A) and (R) are true and (R) is correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is not correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: By Pythagoras theorem, ladder length $= \sqrt{6^2 + 8^2} = \sqrt{100} = 10$ m, so (A) is true. However, $\tan\theta = \dfrac{8}{6} = \dfrac{4}{3} \neq \sqrt{3} = \tan 60°$ ($\theta \approx 53.13° \neq 60°$), so (R) is false.

SECTION B: Very Short Answer Questions MATHS MANTHAN | ($2 \times 2 = 4$ Marks)

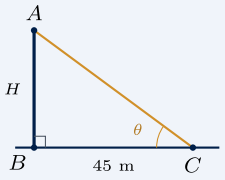
Q6. A vertical pole of height 10 m casts a shadow of 15 m on the ground and at the same time, a tower casts a shadow of 45 m on the ground. Find the height of the tower. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

At the same instant, Sun's elevation θ satisfies $\tan\theta = \dfrac{10}{15} = \dfrac{H}{45}$. [1 Mark]

$$\implies H = \frac{10 \times 45}{15} = \boxed{30\text{ m}}$$

Final tower height: [1 Mark]



Q7. A 1.6 m tall boy stands at a distance of 3 m from a lamp-post and casts a shadow of length 4 m on the ground. Find the height of the lamp-post. **[2]**

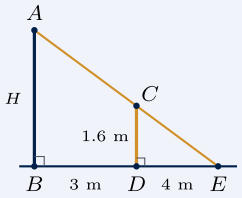
▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

Let lamp-post be $AB = H$ m, boy $CD = 1.6$ m at $BD = 3$ m with shadow $DE = 4$ m. [0.5 Mark]

By similar triangles $\triangle ABE \sim \triangle CDE$: $\dfrac{H}{1.6} = \dfrac{3+4}{4} = \dfrac{7}{4}$. [1 Mark]

$$\implies H = 7 \times 0.4 = \boxed{2.8 \text{ m}}$$

Final lamp-post height: [0.5 Mark]



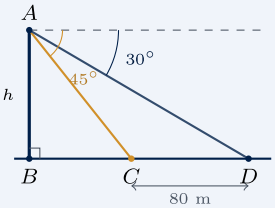
SECTION C: Short Answer Questions — MATHS MANTHAN | (3 × 3 = 9 Marks)

Q8. Two boats are sailing in the sea 80 m apart from each other towards a cliff AB. The angles of depression of the boats from the top of the cliff are 30° and 45° respectively. Find the height of the cliff. (Take $\sqrt{3} = 1.732$) **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let cliff $AB = h$ m and boats on same side be C (45°) and D (30°) with $CD = 80$ m. [0.5 Mark]

In $\triangle ABC$: $BC = h\cot 45° = h$; in $\triangle ABD$: $BD = h\cot 30° = h\sqrt{3}$. [1 Mark]



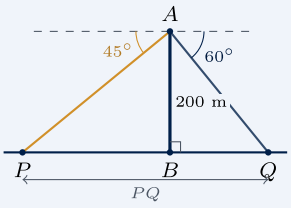
$$CD = h\sqrt{3} - h = 80 \implies h = \frac{80}{\sqrt{3}-1} = \frac{80(\sqrt{3}+1)}{2} = 40(\sqrt{3}+1) = 40(2.732) = \boxed{40(\sqrt{3}+1) \text{ m} \approx 109.28 \text{ m}}.$$ [1.5 Marks]

Q9. An aeroplane at an altitude of 200 metres observes the angles of depression of opposite points on the two banks of a river to be 45° and 60°. Find the width of the river. (Use $\sqrt{3} = 1.732$) **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let aeroplane A be at $AB = 200$ m directly above river line PQ, observing opposite banks P (45°) and Q (60°). [0.5 Mark]

In $\triangle ABP$: $PB = 200\cot 45° = 200$ m. In $\triangle ABQ$: $BQ = \dfrac{200}{\sqrt{3}} = \dfrac{200\sqrt{3}}{3}$ m. [1.5 Marks]



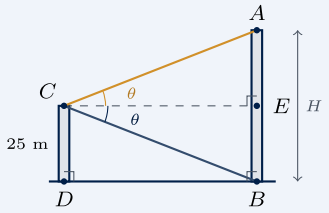
$$\text{Width } PQ = 200 + \frac{200 \times 1.732}{3} = 200 + 115.467 \approx \boxed{200\left(1 + \frac{\sqrt{3}}{3}\right) \text{ m} \approx 315.47 \text{ m}}.$$ [1 Mark]

Q10. From the top of a cliff 25 m high, the angle of elevation of the top of a tower is found to be equal to the angle of depression of the foot of the tower. Find the height of the tower. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let cliff be $CD = 25$ m and tower be $AB = H$ m. Draw $CE \perp AB$ so $BE = CD = 25$ m. Let $\angle ACE = \angle ECB = \theta$ and $CE = x$. [0.5 Mark]

In $\triangle CEB$: $\tan\theta = \dfrac{BE}{x} = \dfrac{25}{x}$. In $\triangle AEC$: $\tan\theta = \dfrac{AE}{x}$. [1.5 Marks]



Comparing both ratios: $\dfrac{AE}{x} = \dfrac{25}{x} \implies AE = 25$ m. Therefore, height of tower $AB = AE + BE = 25 + 25 = \boxed{50 \text{ m}}$. [1 Mark]

SECTION D: Long Answer Question

MATHS MANTHAN | $(1 \times 5 = 5$ Marks)

Note: The source question paper header specifies MM: 25, while the printed section breakdown totals $5 + 4 + 9 + 5 = 23$ Marks.

Q11. The angle of elevation of the top of a tower 24 m high from the foot of another tower in the same plane is 60°. The angle of elevation of the top of the second tower from the foot of the first tower is 30°. Find the distance between the two towers and the height of the other tower. Also, find the length of the wire attached to the tops of both the towers. **[5]**

OR

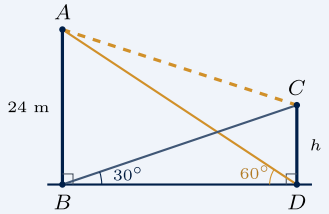
From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 30°. Determine the height of the tower. **[5]**

▶ MAIN QUESTION — STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let towers $AB = 24$ m and $CD = h$ m be $BD = x$ m apart ($\angle ADB = 60°, \angle CBD = 30°$). [1 Mark]

In $\triangle ABD$: $BD = 24\cot 60° = \dfrac{24}{\sqrt{3}} = \boxed{8\sqrt{3} \text{ m} \approx 13.86 \text{ m}}$. [1.5 Marks]

In $\triangle CDB$: $\tan 30° = \dfrac{h}{8\sqrt{3}} \implies h = \boxed{8 \text{ m}}$. [1.5 Marks]



Wire connecting tops A and C (horizontal run $8\sqrt{3}$ m, vertical drop $24 - 8 = 16$ m): [1 Mark]

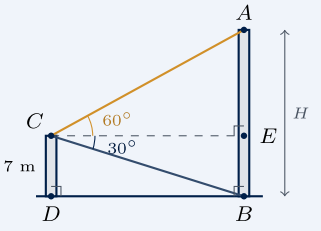
$$AC = \sqrt{(8\sqrt{3})^2 + 16^2} = \sqrt{192 + 256} = \sqrt{448} = \boxed{8\sqrt{7} \text{ m} \approx 21.17 \text{ m}}$$

▶ OR CHOICE — STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let building be $CD = 7$ m and cable tower be $AB = H$ m. Draw $CE \perp AB$ so $BE = 7$ m, $\angle ACE = 60°$, $\angle ECB = \angle CBD = 30°$. [1 Mark]

In right $\triangle CDB$: $\tan 30° = \dfrac{7}{BD} \implies BD = CE = 7\sqrt{3}$ m. [1.5 Marks]

In right $\triangle AEC$: $\tan 60° = \dfrac{AE}{7\sqrt{3}} \implies AE = 7\sqrt{3} \cdot \sqrt{3} = 21$ m. [1.5 Marks]



Total height of cable tower $AB = AE + BE = 21 + 7 = \boxed{28 \text{ m}}$. [1 Mark]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★